

Jayadratha Gayen

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AI Engineer and Researcher with expertise in Large Language Models (LLMs), Agentic AI, Retrieval-Augmented Generation (RAG), and Graph Neural Networks. Experienced in building enterprise-scale AI systems, coding assistants, uncertainty-aware machine learning models, and multilingual AI applications. Published researcher with work spanning trustworthy AI, temporal graph learning, and high-stakes decision-making systems.

Work Experience

CloudAngles Inc.

Feb 2026 – Present

AI Research Engineer

- Building Codebenders, an agentic coding assistant: developed repository indexing, semantic retrieval, and context-management pipelines that facilitate natural language driven code editing, debugging, and generation.
- Cut AI token usage by **76.3%** and API costs by **64.2%** through in-session tool result caching and smarter tool loading, keeping the agent cost-efficient at scale.
- Improved response speed by **22.4%** with better context and memory management, allowing the agent to resume exactly where it left off after any interruption.
- Built automated **LLM evaluation pipelines** for benchmarking code quality, task reliability, and model performance across real developer scenarios.

Google Summer of Code at INCF

May 2025 – Sep 2025

Open Source Developer

- Developed **DevoTG**, a temporal graph neural network framework for modeling *C. elegans* embryogenesis and connectome formation.
- Modeled spatiotemporal cell division and neuronal connectivity as **dynamic graph datasets** using PyTorch and PyG.
- Built interactive **3D graph visualizations** for biological data exploration.

Vikram Solar Limited

Jul 2018 – Dec 2019

Engineer

- Did **Six Sigma** throughput improvement project increasing production by **10%**.
- Performed bottleneck analysis using **DOE, Pareto, Five Why, Fishbone**.
- Implemented **RFID automation** and data analytics workflows using SAP and Excel.

Technical Skills

Programming: Python, C, C++, SQL

ML/AI: PyTorch, TensorFlow, PyG, Transformers, LLMs, Agentic AI, RAG, AI Coding Agents, LLM Evaluation

Frameworks: LangGraph, LangChain, HuggingFace Transformers, Model Context Protocol (MCP)

Infrastructure: AWS, Docker, Git, MongoDB, Neo4j, Knowledge Graphs, Vector Databases

Education

International Institute of Information Technology Hyderabad

M.Sc in Electronics and Communication Engineering by Research (CGPA: 8.29/10)

2022 – 2025

Indian Institute of Engineering Science and Technology Shibpur

Bachelor of Technology in Electrical Engineering (CGPA: 8.02/10)

2014 – 2018

Selected Publications

Node Classification With Reject Option (TMLR 2025)

- Proposed **NodeCwR**, a reject-option GNN framework that improves prediction reliability by abstaining on uncertain cases in high-stakes applications like legal, medical, etc.

Reducing Misclassification Risk in Dynamic Graph Neural Networks Through Abstention (ASONAM 2025)

- Introduced abstention mechanisms for temporal GNNs, improving AUC/AP by **10–15%** while addressing class imbalance and critical misclassifications.

DevoTG: Temporal Graph Neural Networks for Modeling *C. elegans* Developmental Connectomics (arXiv 2026)

- Built a temporal GNN framework for modeling how *C. elegans* nervous system develops; temporal approach achieved mean test AUC of 0.839 vs 0.577 for static methods.

Selected Projects

Nyay Saheli: Conversational Legal Assistant (LLM + GraphRAG)

2025

- Built a multilingual legal AI assistant using **RAG** and **GraphRAG** for legal guidance and question answering.
- Developed OCR, ingestion, embedding, hybrid-search, reranking, and retrieval pipelines over legal knowledge bases.
- Utilized vector databases and knowledge graphs for context-grounded reasoning and response generation.
- Designed culturally aware, empathetic, multilingual conversational workflows for Indian legal and social contexts.

Multilingual News Similarity Detection (Transformers)

2022

- Developed a **semantic document similarity system** using Siamese Transformer architectures.
- Implemented **DistilBERT-based multilingual embeddings** for cross-language document retrieval.
- Evaluated TF-IDF, LSTM, and Transformer models achieving best performance with DistilBERT.

Node Classification with Integrated Reject Option for Legal Judgement Prediction

2024

- Introduced a GNN framework, NodeCwR, with cost-based and coverage-based rejection to improve the reliability of Legal Judgment Prediction (LJP) by analysing **complex legal texts** and disease prediction tasks.
- Achieved 6-19% improvement in accuracy on the ILDC dataset for LJP task after abstaining from uncertain cases.
- Enhanced interpretability of the rejection decisions using SHAP visualizations.

Leadership & Activities

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- National Finalist (Top 6/275 teams), **BharatGen Hackathon** – GenAI legal assistant.
 - Teaching Assistant, **3D Vision Summer School**, IIIT Hyderabad.
 - Founding Member, **Srijani – The Hope** education initiative.